

ODESSA-SCHLEMEYER FIELD, TX, USA

WMO: 722648

Lat: 31.921N

Lon: 102.387W

Elev: 3001

StdP: 13.17

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03031

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.3	(c) 25.3	(d) 0.6	(e) 6.3	(f) 44.9	(g) 4.2	(h) 7.6	(i) 48.0	(j) 28.6	(k) 52.7	(l) 26.3	(m) 49.9	(n) 8.3	(o) 40	(p) 0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	100.7	65.8	98.7	66.5	96.6	66.6	72.3	87.2	71.4	86.5	70.7	85.7	11.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.0	119.2	75.7	67.8	114.3	75.5	66.6	109.5	75.8	37.9	87.5	37.1	86.4	36.5	85.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.6	(b) 22.8	(c) 20.1	(e) 16.3	(f) 105.6	(g) 4.0	(h) 2.8	(i) 13.5	(j) 107.6	(k) 11.1	(l) 109.3	(m) 8.9	(n) 110.8	(o) 6.0	(p) 112.9		
(a) 25.6	(b) 22.8	(c) 20.1	(e) 14.0	(f) 74.1	(g) 3.6	(h) 1.4	(i) 11.5	(j) 75.1	(k) 9.4	(l) 75.9	(m) 7.4	(n) 76.6	(o) 4.8	(p) 77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.5	45.7	49.6	58.6	66.5	74.2	82.3	83.0	82.8	75.6	66.0	54.5	46.1	(6)
(7)		DBStd	15.78	8.91	10.48	9.37	8.25	7.86	4.99	4.43	4.39	5.92	8.49	9.10	8.65	(7)
(8)		HDD50	595	191	123	33	4	1	0	0	0	0	7	59	177	(8)
(9)		HDD65	2379	599	436	238	80	20	0	0	0	4	90	327	585	(9)
(10)		CDD50	6247	57	112	298	499	750	970	1023	1016	767	504	194	57	(10)
(11)		CDD65	2555	0	4	38	125	304	520	558	551	321	122	12	0	(11)
(12)		CDH74	30610	18	119	562	1557	3589	6527	6925	6779	3182	1173	155	24	(12)
(13)		CDH80	15560	1	22	149	615	1818	3648	3780	3682	1414	405	26	0	(13)
(14)	Wind	WSAvg	10.5	9.3	10.4	11.4	12.3	12.2	12.1	10.7	9.8	9.6	9.8	9.4	9.3	(14)
(15)	Precipitation	PrecAvg	12.9	0.6	0.7	0.7	0.7	1.4	1.3	1.7	1.8	1.4	1.3	0.7	0.5	(15)
(16)		PrecMax	23.1	1.7	2.2	2.0	2.3	7.1	3.1	6.2	4.6	4.5	4.1	4.9	2.7	(16)
(17)		PrecMin	5.6	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.5	0.6	0.6	0.5	0.7	1.7	0.9	1.6	1.2	1.2	1.3	1.0	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	83.5	88.6	93.4	101.3	105.1	102.3	101.9	97.5	91.5	83.7	77.5	(19)
(20)			MCWB	51.0	54.7	55.7	57.4	61.6	63.8	67.6	67.6	66.5	62.5	56.8	51.7	(20)
(21)		2%	DB	71.9	77.4	84.3	90.5	97.3	101.5	99.3	99.2	93.8	88.2	78.6	71.9	(21)
(22)			MCWB	49.6	51.2	55.1	57.0	60.7	64.4	67.9	68.0	66.2	62.7	56.9	50.2	(22)
(23)		5%	DB	66.8	73.1	80.6	87.1	93.5	98.5	97.2	97.1	91.1	84.8	74.4	66.3	(23)
(24)			MCWB	47.4	50.6	54.0	56.8	60.7	65.9	67.8	68.1	65.9	62.4	54.8	48.4	(24)
(25)		10%	DB	62.2	68.3	76.4	83.2	90.3	95.2	94.7	94.5	88.2	81.1	70.5	61.8	(25)
(26)			MCWB	46.4	48.6	53.3	56.9	61.1	66.9	67.9	68.1	66.1	61.2	53.4	47.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	59.1	63.9	66.9	70.8	73.3	73.4	73.2	72.2	70.8	64.6	58.0	(27)
(28)			MCD B	63.5	68.8	73.9	79.4	84.4	89.0	87.2	88.4	85.6	81.5	70.8	64.2	(28)
(29)		2%	WB	53.1	56.3	61.6	65.0	69.4	72.0	72.0	72.1	70.8	68.7	61.2	53.9	(29)
(30)			MCD B	62.8	67.3	70.6	76.0	81.4	87.0	86.7	88.0	83.9	78.3	69.9	63.8	(30)
(31)		5%	WB	50.0	53.3	59.7	63.3	68.2	71.0	71.1	71.2	69.9	67.2	59.1	51.0	(31)
(32)			MCD B	64.0	68.0	69.8	74.7	80.0	85.9	86.2	87.2	82.2	76.2	67.9	63.2	(32)
(33)		10%	WB	47.1	50.7	57.3	61.4	66.9	70.0	70.3	70.3	68.9	65.4	56.7	48.2	(33)
(34)			MCD B	61.5	66.0	69.4	73.1	79.0	85.4	85.5	86.0	80.8	74.9	66.7	61.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	24.8	25.6	26.5	26.6	24.6	23.8	21.3	21.5	20.9	23.1	24.1	23.9	(35)
(36)			MCDBR	32.0	32.5	30.9	30.1	29.2	27.2	24.0	23.9	24.6	26.5	28.5	31.3	(36)
(37)			MCWBR	15.5	14.5	12.3	12.0	11.1	7.9	6.3	6.0	7.3	10.0	12.3	15.4	(37)
(38)		5% WB	MCD BR	27.5	27.8	24.7	24.6	24.4	23.3	20.9	21.1	19.3	19.9	21.9	26.4	(38)
(39)			MCWBR	14.9	14.4	12.1	12.1	11.1	7.5	6.0	5.9	6.5	9.4	11.8	14.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.263	0.273	0.292	0.326	0.346	0.375	0.395	0.370	0.350	0.311	0.284	0.268	(40)
(41)		taud		2.582	2.555	2.516	2.408	2.417	2.384	2.371	2.416	2.440	2.540	2.546	2.566	(41)
(42)		Ebn at Noon		311	316	315	305	298	288	281	288	291	298	299	303	(42)
(43)		Edh at Noon		25	29	32	37	37	38	39	36	34	29	26	24	(43)
(44)	All-Sky Solar Radiation	RadAvg		1107	1373	1775	2169	2307	2422	2296	2130	1780	1460	1162	991	(44)
(45)		RadStd		106	139	127	133	154	110	124	92	120	156	120	87	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a) Station Only	(b) Regional (15 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (15 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page